# Call associations before and after August 2024

By Marina Levy

**Background:**

I chose this research question because I am interested in the integration of MCSLC (Mobile Crisis Service of Lane County) into Eugene/Lane county and how that affected CAHOOTS (Crisis Assistance Helping Out On The Streets) in the aftermath. My major goals for this project are to determine if the call types changed for CAHOOTS and EPD (Eugene Police Department) before and after the introduction of MCSLC. In this reality where CAHOOTS no longer exist, it's very important to show using data the importance of CAHOOTS and the work they do. CAHOOTS plays such an important role in our community and it is important to me that we show that through the data and using evidence. CAHOOTS provided an alternative response to non violent calls where a trained mental health professional and a trained EMT were deployed to help individuals without the police being involved where the situation can easily escalate with EPD's presence.

**Data:**

The data I am using is the CAD data that ranges from 2014 to 2025 in order to see the effects that the introduction of MCSLC had on CAHOOTS. This data was provided to the class specifically for this project. I looked at all the observations where the 'primeunit' was clearly stated. The variables I looked at were: 'nature', 'primeunit', my own 'date' column and 'yr'. Some important strengths in the data were calls that were answered by CAHOOTS were labeled as such and the sheer amount of observations that were used to get an accurate result. Some limitations and caveats were the inability to identify if calls were responded to by MCSLC and the variables in the 'nature' not being as specific as I would have liked in order to fully understand the types of calls that are being responded to by both CAHOOTS and other organizations.

The steps I took once I had the data were as follows:

1. I downloaded the csv files from canvas
2. I loaded in each file because they were separated by year
3. Once I had them all I made a list
4. Then I used pd.contact() to make them into one data frame.

**Methods**

The packages I used were:

- Numpy
- Pandas
- Matplotlib.pyplot
- Matplotlib.dates
- Seaborn
- Scipy stats

The steps I took to clean the data were:

- I read in every data frame per year in jupiter notebook

- I made a list of all the data frames and used pd.concat() to make them into one data frame
- I used pd.to_datetime() and dt.time and dt.date to turn the calltime column to into a time variable and made a new column called date with the date column
- I dropped columns that were not needed
- I dropped NaN values in the primeunit column
- I separated the data into cahoots and not cahoots in the prime unit column using .str.startswith('CAHOOT')
- I made a cutoff date pd.Timestamp('2024-08-01') and used it to separate the data into before and after the cutoff.
- Finally, I combined all the data sets back into one using .contact()

Analytical steps:

- I loaded in the test I'm going to use: from scipy.stats import chi2_contingency
- I found the 20 most common in the nature column and decided to perform a Chi squared test on these 20 the output was a list.
- Next I need to start the Chi squared test using chi2_contingency finding the chi2 and p values
- Next I plotted the proportion of each call in a bar chart and a barchart of the p-values using matplotlib and seaborn.

**Results:**

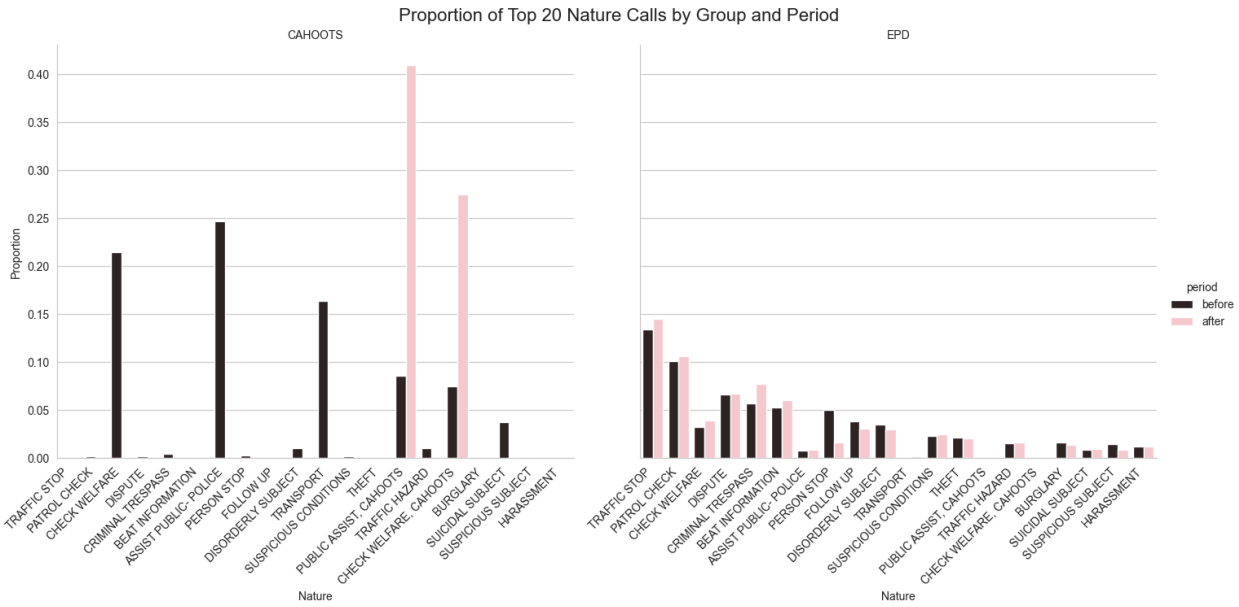
The first result I made was an individual report of each of the 20 most common reasons for a call. I chose 20 because of the amount of reasons for a response. By taking the sum of all the unique values for the 'nature' column and selecting the largest. After taking the 20 I used a for loop to get the results of each Chi squared test. Each report has the name of the call type, a table of the amount of calls for CAHOOTS, EPD by before and after the cutoff of August 2024, the Chi square statistic, p-value and whether it was statistically significant.

| | | |
|---|---|---|
| Nature: TRAFFIC STOP<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 29<br>EPD 8889 118849<br>Chi-square statistic:<br>1.2271<br>P-value: 0.2680<br>Not statistically<br>significant | Nature: PATROL CHECK<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 2 269<br>EPD 6523 89785<br>Chi-square statistic:<br>14.6807<br>P-value: 0.0001<br>Statistically<br>significant | Nature: CHECK WELFARE<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 7 36776<br>EPD 2410 28845<br>Chi-square statistic:<br>2915.5010<br>P-value: 0.0000<br>Statistically<br>significant |
| Nature: DISPUTE<br>Contingency Table:<br>period after before<br>group | Nature: CRIMINAL<br>TRESPASS<br>Contingency Table:<br>period after before | Nature: BEAT INFORMATION<br>Contingency Table:<br>period after before<br>group |

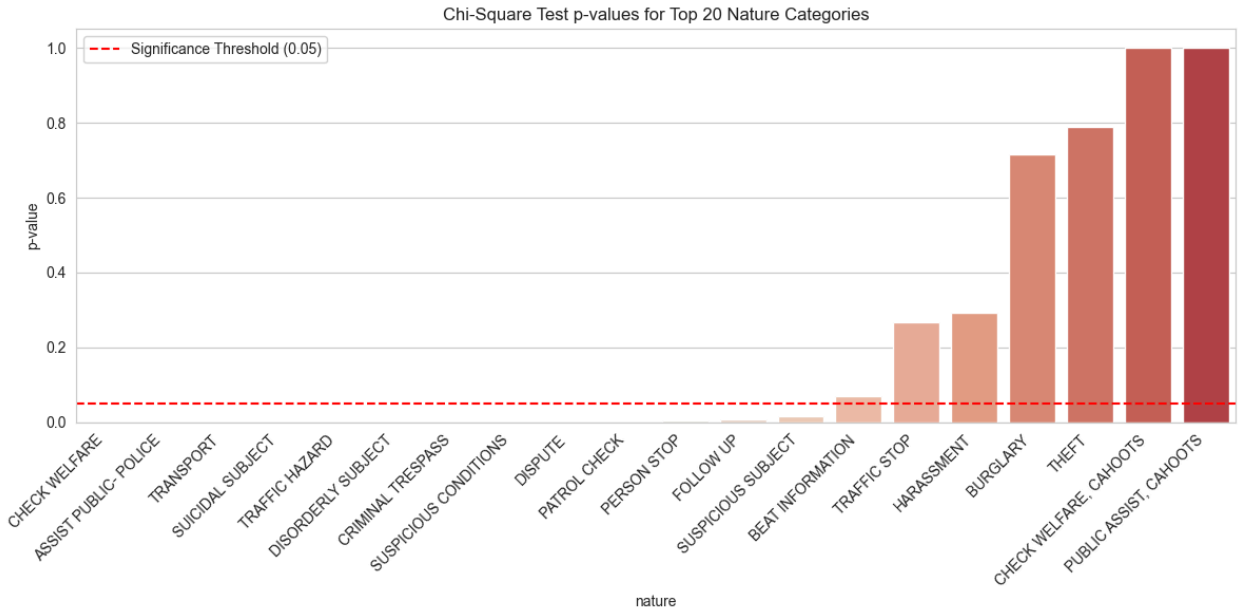
| | | |
|---|---|---|
| CAHOOTS 0 248<br>EPD 4115 58795<br>Chi-square statistic: 16.2952<br>P-value: 0.0001<br>Statistically significant | group<br>CAHOOTS 0 660<br>EPD 4716 50420<br>Chi-square statistic: 60.5640<br>P-value: 0.0000<br>Statistically significant | CAHOOTS 1 76<br>EPD 3687 46808<br>Chi-square statistic: 3.2582<br>P-value: 0.0711<br>Not statistically significant |
| Nature: ASSIST PUBLIC-POLICE<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 1 42206<br>EPD 523 6848<br>Chi-square statistic: 3012.1623<br>P-value: 0.0000<br>Statistically significant | Nature: PERSON STOP<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 425<br>EPD 974 44573<br>Chi-square statistic: 8.2826<br>P-value: 0.0040<br>Statistically significant | Nature: FOLLOW UP<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 147<br>EPD 1860 33671<br>Chi-square statistic: 7.0937<br>P-value: 0.0077<br>Statistically significant |
| Nature: DISORDERLY SUBJECT<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 1784<br>EPD 1809 30538<br>Chi-square statistic: 104.2425<br>P-value: 0.0000<br>Statistically significant | Nature: TRANSPORT<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 27947<br>EPD 73 969<br>Chi-square statistic: 1935.0504<br>P-value: 0.0000<br>Statistically significant | Nature: SUSPICIOUS CONDITIONS<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 310<br>EPD 1500 20295<br>Chi-square statistic: 21.8132<br>P-value: 0.0000<br>Statistically significant |
| Nature: THEFT<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 12<br>EPD 1221 19087<br>Chi-square statistic: 0.0721<br>P-value: 0.7882<br>Not statistically significant | Nature: PUBLIC ASSIST, CAHOOTS<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 4558 14614<br>EPD 0 2<br>Chi-square statistic: 0.0000<br>P-value: 1.0000<br>Not statistically significant | Nature: TRAFFIC HAZARD<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 1721<br>EPD 999 13710<br>Chi-square statistic: 123.2669<br>P-value: 0.0000<br>Statistically significant |

| | | |
|---|---|---|
| Nature: CHECK WELFARE, CAHOOTS<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 3051 12740<br>EPD 0 2<br>Chi-square statistic: 0.0000<br>P-value: 1.0000<br>Not statistically significant | Nature: BURGLARY<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 15<br>EPD 840 14532<br>Chi-square statistic: 0.1315<br>P-value: 0.7169<br>Not statistically significant | Nature: SUICIDAL SUBJECT<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 4 6424<br>EPD 560 7700<br>Chi-square statistic: 439.9448<br>P-value: 0.0000<br>Statistically significant |
| Nature: SUSPICIOUS SUBJECT<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 159<br>EPD 537 12858<br>Chi-square statistic: 5.6254<br>P-value: 0.0177<br>Statistically significant | Nature: HARASSMENT<br>Contingency Table:<br>period after before<br>group<br>CAHOOTS 0 29<br>EPD 738 10517<br>Chi-square statistic: 1.1033<br>P-value: 0.2935<br>Not statistically significant | |

The second visualization I made from my results is the following bar chart showing the proportion of calls before and after the cutoff for CAHOOTS and EPD. After running the Chi squared test I found the proportion and plotted their frequency by taking each element and dividing the frequency by the total.


Proportion of Top 20 Nature Calls by Group and Period
CAHOOTS
EPD
Proportion
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
TRAFFIC STOP
PATROL CHECK
CHECK WELFARE
DISPUTE
CRIMINAL TRESPASS
BEAT INFORMATION
ASSIST PUBLIC- POLICE
PERSON STOP
FOLLOW UP
DISORDERLY SUBJECT
TRANSPORT
SUSPICIOUS CONDITIONS
THEFT
PUBLIC ASSIST. CAHOOTS
TRAFFIC HAZARD
CHECK WELFARE. CAHOOTS
BURGLARY
SUICIDAL SUBJECT
SUSPICIOUS SUBJECT
HARASSMENT
Nature
period
before
after

The final visualization I made was the following bar chart of the p-values after conducting my Chi squared test. I organized the p-values from smallest to largest. I included the threshold cutoff of $p < 0.05$ to show if the different call types were statistically significant in a more cohesive and digestible manner.




**Discussion:**

The results I produced did answer my research question of whether the introduction of MCSLC in august of 2024 changed the call types of CAHOOTS or EPD. My results show that the change in call types for CAHOOTS after August 2024 when MCSLC was introduced by selecting the most recurring call reasons, I determined the changes were not statistically significant from the p-values. Meaning, the introduction of MCSLC did not change the way CAHOOTS or EPD operated. One caveat of my results are Check welfare vs Check welfare, CAHOOTS and Public assist vs Public assist, CAHOOTS values. Having the difference in the two column types and their frequency changes the results of the test. However, after further investigation, the difference in those call types would not have changed the results of the test. Some limitations are the confusing observations in the primeunit column making it hard to identify who is being deployed and there is no way of knowing when MCSLC was deployed. A weakness in my results is that I would have liked to perform another statistical test on the data to see if the statistical significance would change in order to know if the Chi squared test had some limitations. Additionally, I should have removed any observations that had the ‘callsource’ as self because that means EPD took the action without being deployed, something CAHOOTS does not do. The next steps for me would be looking at call priority before and after the change. I think it would be interesting to see if the introduction of MCSLC changed the call priority for CAHOOTS or EPD and how that related to the different variables of the ‘nature’column of each call.